Project Overview

SOZO

Project Overview

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Prepared By: SOZO

Client: John Quentin / Choose My Lawyer Ltd

Choose My LawyerSRA API Data Extraction

The source data for all UK Law Firms is from the SRA. We have extracted and reviewed the data from their API and it appears suitable for CML requirements.

We have extracted and reviewed Law Firm data from the SRA API and it seems suitable for CML requirements.

https://www.sra.org.uk/sra/how-we-work/privacy-data-information/data-sharing/

sra_data_15_oct_25

Statistics

- Total organisations: 24,973
- Authorised (YES): 9,020
- Not authorised (null): 13,937
- Ceased: 494
- Revoked: 1,371
- Intervened: 142
- Conditions: 9

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Scraping Workflow

Scrape websites of Authorised firms to attempt to extract Pricing Transparency Statements (PTS) to map into website's required pricing data.

1. Check for website existence
2. Discover sitemaps from the website (check robots.txt for reference)
3. Keyword based URL filtering and weighting
4. Scrape the filtered URLs
5. Process with AI and extract useful information
6. Plan how often we would check the SRA API for new or updated entries

Unknowns

1. Finding out from 9020 Authorised firms, how many of them have a website. (7357)
2. If they have a website, is it working or dead.
3. If it's working then do they have a pricing transparency page listed?

Technical Stack Assumptions

- Python (FastAPI) - Backend (Better AI/ML library support)
  Postgres - DB (Strong JSONB support for flexible schemas)
- Redis - Caching layer
- React (Next.js) - Frontend (SEO requirements, server-side rendering)
- Vector DB - AI Embeddings Pinecone
- WordPress - Frontend Website

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External Services

- SRA Data API
- Email Provider - Sparkpost

External Services Assumptions

- AI/LLM Provider - Claude / OpenAI
- Scraping - Jina.ai
- Reviews - https://www.reviewsolicitors.co.uk/ (Phase II)

Ranking Factors

- Price (60% weight): Lower prices rank higher
- Reputation (25% weight): Star rating from reviews
- Distance (15% weight): Proximity to user location

Monetisation

- Start with free for all so you can appoint any firm
- Referral fee for an appointment at some point
- Will need to link up to Xero or other to generate invoices to send and chase up

Phase I - MVP

- Wills Trust and Probate
- Conveyancing

Functional Specifications

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Functional Specification & Executive Summary

Table of Contents

1. Executive Summary
2. Project Overview
3. User Personas
4. Consumer User Flows
5. Law Firm User Flows
6. Feature Specifications
7. Review System
8. Email Notifications
9. Content Requirements
10. Business Rules

1. Executive Summary

1.1 Project Vision

Choose My Lawyer (CML) is the first genuine whole-of-market digital comparison tool for solicitors in England and Wales. The platform enables consumers to instantly compare, select, and appoint law firms based on price, reputation, location, practice area, and regulatory history.

1.2 Problem Statement

Consumers of legal services currently lack effective tools to compare solicitors. Existing options are limited to:

- Asking friends and family for recommendations
- Using a solicitor they've used before (regardless of quality)
- Searching Google for the nearest option
- Using directory sites that provide leads but no actual prices

There is no platform offering transparent, comparable pricing data across the legal services market.

1.3 Solution

CML solves this by:

- **Extracting pricing data** from SRA-mandated price transparency statements using AI
- **Standardising price information** into comparable formats
- **Providing instant quotes** through a conversational interface
- **Enabling direct appointment** of solicitors through the platform

1.4 Unique Selling Propositions

Feature	CML	Competitors
Whole-of-market comparison	✓	✗
Instant price comparison	✓	✗
Instant instruction/appointmen	✓	✗
AI-powered chat interface	✓	✗
Actual calculated prices	✓	Lead generation only
Appoint Law Firm	✓	✗

1.5 Law Firm Target Market

- **Primary:** 7,200 small to medium firms in England & Wales serving B2C/B2H
- **Market Size:** £15 billion annual turnover in target practice areas
- **Key Practice Areas:** Residential conveyancing (£3.81bn), Wills/Trusts/Probate (£2.68bn), Family (£2.41bn), Employment (£1.79bn), Personal Injury (£1.75bn)

1.6 Revenue Model

- **Referral fee:** £125 per confirmed instruction
- **Monetisation triggers:**
    - Direct appointment through platform
    - Callback request leading to instruction

- **Future:** Subscription tiers for enhanced firm features

1.7 MVP Scope

The initial release focuses on **two practice areas**:

1. **Residential Conveyancing** (purchases, sales, remortgages, transfers of equity)
2. **Probate** (uncontested, grant-only and full administration)

2. Project Overview

2.1 Platform Architecture

The solution comprises two distinct platforms:

Platform 1: WordPress Content Site

- Marketing content and landing pages
- Blog and knowledge articles
- SEO-optimised practice area pages
- Static informational content
- Entry points to the comparison tool

Platform 2: Custom Web Application

- Conversational chat interface
- Price comparison engine
- Results display and sorting
- Appointment/callback functionality
- Law firm dashboard and registration
- Administrative functions

2.2 Key Stakeholders

Stakeholder	Role	Primary Goals
Service Users (Consumers)	End users seeking legal services	Find best suited solicitor quickly
Law Firms	Service providers	Receive qualified instruction
CML (Platform)	Marketplace operator	Generate referral revenue
SRA	Regulator	Ensure price transparency compliance

2.3 Success Metrics

- **User Metrics:** Chat completions, quote generations, appointment conversions
- **Firm Metrics:** Enrolled firms, pricing data coverage, response rates
- **Revenue Metrics:** Appointments generated, referral fees collected
- **Quality Metrics:** User satisfaction, review scores, price accuracy

3. User Personas

3.1 Consumer Personas examples

Persona 1: First-Time Buyer (Sarah, 28)

- Buying first home, unfamiliar with conveyancing process
- Price-conscious but values reputation
- Prefers mobile-friendly, guided experience
- Wants to understand what's included in quotes

Persona 2: Executor (David, 55)

- Recently lost parent, needs to handle probate
- Emotionally stressed, wants straightforward process
- Values experience and clear communication
- Less price-sensitive, prioritises trust

Persona 3: Property Investor (James, 42)

- Multiple property transactions per year
- Highly price-sensitive, knows what to expect
- Values speed and efficiency
- Wants to compare multiple quotes quickly

3.2 Law Firm Personas examples

Persona 1: Small High Street Firm

- 2-5 solicitors, local client base
- Limited marketing budget
- Wants low-cost lead generation
- Needs simple dashboard

Persona 2: Regional Specialist

- 10-30 solicitors, conveyancing focus
- Competitive pricing, volume-based model
- Wants visibility in search results
- Interested in analytics and optimisation

4. Consumer User Flows

4.1 Entry Points

Consumers can enter the comparison journey from:

1. **Homepage** - Central search/chat widget
2. **Practice Area Landing Pages** - Service-specific CTAs (e.g., "Compare Conveyancing Quotes")
3. **Direct Link** - Saved session URL from email
4. **Get Suggestions** - Service selector dropdown

4.2 Chat Flow Overview

The chat interface presents questions sequentially, with users primarily selecting options via buttons rather than typing. Free-text input is supported but not required.

4.3 Conveyancing Chat Flow

Step	Question	Input Type	Options
1	Natural language or service selection	Text/Selection	"I'm moving house...", "Conveyancing"
2	What are you doing?	Single select	Sale & Purchase, Purchase Only, Sale Only, Remortgage
3	Is the property freehold or leasehold?	Single select	Freehold, Leasehold, Not sure
4	What is the property value?	Number input	£ amount
5	What is the property address?	Text input	Postcode/address
6	Additional factors	Multi-select checkboxes	Mortgage required, New build, Part of chain
7	Any purchase schemes?	Multi-select	Help to Buy ISA, Shared Ownership, None
8	Location preference for solicitor	Text input	County/region or blank for remote
9	How should we rank results?	Single select	Best Reviewed, Price, Location

4.4 Probate Chat Flow

Step	Question	Input Type	Options
1	Service identification	Text/Selection	"Probate", "Someone has died..."
2	What service do you need?	Single select	Grant only, Full administration
3	Estimated gross estate value	Number input	£ amount
4	Is there a valid Will?	Single select	Yes (testate), No (intestate)

Step	Question	Input Type	Options
5	Is the estate taxable (IHT400)?	Boolean	Yes, No
6	Was the deceased UK domiciled?	Boolean	Yes, No
7	Number of UK properties	Number input	Count
8	Number of bank accounts	Number input	Count
9	Number of investments/shareholdings	Number input	Count
10	Any assets outside the UK?	Boolean	Yes, No
11	Number of beneficiaries	Number input	Count
12	Location preference	Text input	County/region or blank
13	How should we rank results?	Single select	Best Reviewed, Price, Location

4.5 Progress Indicator

- Left sidebar displays answered questions summary
- Progress bar shows approximate completion (simplified, not exact percentage)
- Progress may hold at same level for 2-3 questions before advancing

4.6 Results Display

Top 5 Results Table (Enrolled Firms)

Column	Description
Rank	Position 1-5
Firm Name	Practice name with service tags
Reputation	Star rating + review count (e.g., "4.8 / 5.0 (1024 reviews)")
Total Cost	Calculated quote based on answers
Distance	Miles from user's location
Number of Offices	Office count
Regulatory Status	Green/Amber/Red indicator

Column	Description
Actions	"Appoint" and "Request a callback" buttons

Extended Results (Position 6+)

- Shows non-enrolled firms with basic data
- No action buttons (greyed out)
- Indicates "Not enrolled - contact directly"

Sorting Options

- Best Reviewed (default based on user selection)
- Price (lowest to highest)
- Location (nearest first)

4.7 Appointment Flow

Direct Appointment

1. User clicks "Appoint" on chosen firm
2. Confirmation modal displays company name, work type, and quoted price
3. User confirms via checkboxes that they want to appoint and share contact details
4. User enters full name and email address
5. User clicks "Proceed to final step" or "Request a call back" or "Cancel"
6. System sends email to firm with client details and confirmation email to user

Callback Request

1. User clicks "Request a call back"
2. Modal collects full name, email, phone number, and preferred contact time
3. User clicks "Request call back"
4. System sends callback request to firm and confirmation email to user

4.8 Session Management

Save for Later

- User clicks "Save for later"

- Modal collects full name and email
- System sends unique URL to resume session
- Session valid for 30 days

Start New Chat

- User clicks "Start new chat"
- Warning modal displayed with option to save current session first
- Confirmation before clearing

No User Accounts

- Consumers operate in guest mode
- No login/registration required
- Session persistence via unique URLs only

5. Law Firm User Flows

5.1 Enrollment Process

1. CML identifies firm from SRA data
2. CML contacts firm with unique enrollment link
3. Firm clicks link → lands on pre-populated profile page
4. Firm reviews/edits details
5. Firm creates account (email, password)
6. Firm accepts terms and conditions
7. Firm reviews/updates pricing data
8. Account activated

5.2 Dashboard Overview

Dashboard Home

- Search appearances (last 30 days)
- Click-through rate
- Appointments received
- Callback requests

- Review score summary

Profile Management

- Firm details (name, contact info)
- Office locations
- Practice areas offered
- Logo/branding

Pricing Management

- View current price cards
- Update pricing data
- Add/remove services

Appointments

- List of received appointments
- Status tracking (pending, confirmed, completed)
- Client contact details

Reviews

- View all reviews (CML + aggregated)
- Respond to reviews
- Cannot delete reviews

5.3 Review Response Flow

1. Firm receives email notification of new review
2. Firm logs into dashboard
3. Navigates to Reviews section
4. Clicks "Respond" on review
5. Enters response text (max 500 characters)
6. Response appears publicly below review

6. Feature Specifications

6.1 Chat Interface

Input Methods

- Button/chip selection (primary)
- Text input field
- Voice input (Web Speech API)

AI Capabilities

- Intent recognition from natural language
- Entity extraction (amounts, addresses, dates)
- Address/postcode lookup and validation
- Clarification questions for ambiguous input

Visual Elements

- Message bubbles (user right-aligned in teal, system left-aligned in white)
- Selection buttons within messages
- Progress sidebar on left
- Input field at bottom with send button and voice icon

6.2 Price Calculation Engine

Process

1. Match user answers to firm price cards
2. Identify applicable pricing band
3. Calculate base fee
4. Apply adjustments (supplements, discounts)
5. Add estimated disbursements
6. Calculate VAT where applicable
7. Return itemised quote

Output Format

- Legal Fees

- Adjustments (with descriptions)
- Disbursements (itemised)
- VAT (on fees)
- Total Estimate

6.3 Results Sorting & Filtering

Sort Options

- Best Reputation (weighted aggregate rating)
- Price (total cost, lowest first)
- Location (distance from user, nearest first)

Display Logic

- Enrolled firms: Positions 1-5 with full functionality
- Non-enrolled firms: Position 6+ with limited display
- Firms without pricing: Shown but marked "Price not available"

6.4 Regulatory Status Indicator

The regulatory status is derived from data provided by the SRA Data Share API. The SRA publishes each firm's authorisation status, any conditions placed on their practice, and historical regulatory interventions. CML maps these statuses to a simple traffic light indicator:

Colour	Meaning	SRA Status
Green	Good standing	Authorised with no conditions or recent interventions
Amber	Caution	Authorised but with conditions, undertakings, or recent warnings
Red	Issues	Licence suspended, under intervention, or authorisation revoked

7. Review System

7.1 Review Sources

Reviews are aggregated from multiple platforms to provide comprehensive reputation data:

Source	Priority	Integration
CML Platform	1 (Highest)	Direct submission via email link
Google Reviews	2	Google Places API
Facebook	3	Facebook Graph API
Trustpilot	4	Trustpilot API

7.2 Review Aggregation

For each enrolled firm, CML discovers and links their external business profiles (Google Place ID, Facebook Page ID, Trustpilot business ID). A scheduled sync process fetches reviews from these platforms periodically to keep data current.

Reviews from all sources are stored locally with source attribution, enabling display even when external APIs are unavailable.

7.3 CML Native Reviews

Collection Process

1. User completes legal matter (3-6 months after appointment)
2. System sends review invitation email
3. Email contains unique token link
4. User clicks link → review form
5. User submits rating (1-5 stars) and text review
6. Review marked as "Verified" on platform

Verification

- Only users with completed appointments can leave CML reviews
- Token prevents fraudulent submissions
- Reviews linked to specific appointment record

7.4 Rating Calculation

Aggregate ratings use a weighted average that prioritises verified CML reviews:

- CML reviews weighted 2x (verified transactions)
- External reviews (Google, Facebook, Trustpilot) weighted equally at 1x
- Display format: "4.8 / 5.0 (1,024 reviews)"
- Total count shown is sum across all sources
- Source breakdown available when user clicks on rating

7.5 Firm Review Response

- Firms can respond to any review via their dashboard
- Responses appear publicly below the review
- Firms cannot edit or delete reviews
- Response notification sent to reviewer (CML reviews only)
- Firm responses limited to 500 characters

7.6 Review Moderation

- Automated checks filter profanity and spam
- Reviews flagged by automated checks go to manual moderation queue
- Firms can report inappropriate reviews for admin review
- Moderation decisions are logged for audit purposes

8. Email Notifications

8.1 Consumer Emails

Trigger	**Subject**	**Content**
Session saved	"Your Choose My Lawyer quote - saved"	Unique URL to resume, session summary
Appointment created	"Appointment confirmed with [Firm]"	Firm details, quoted price, next steps, support contact

Trigger	Subject	Content
Callback requested	"Callback request sent to [Firm]"	Confirmation, expected response time
Review invitation	"How was your experience with [Firm]?"	Link to leave review (sent 3-6 months after appointment)

8.2 Firm Emails

Trigger	Subject	Content
New appointment	"New client appointment via CML	Client details, matter type, quoted price, proforma
Callback request	"Callback request via CML"	Client details, phone number, preferred time
New review	"New review on Choose My Lawye	Review content, rating, link to respond
Weekly digest	"Your CML weekly summary"	Search appearances, leads, rating changes

9. Content Requirements

9.1 WordPress Site Pages

Core Pages

- Homepage
- How It Works
- About Us
- Contact Us
- Privacy Policy
- Terms of Service
- Cookie Policy

Practice Area Landing Pages

- Conveyancing
- Probate

For Law Firms

- Benefits of joining
- How it works for firms
- Pricing/fees
- Registration CTA

Knowledge Hub

- Blog articles
- Guides (e.g., "What to expect when buying a house")
- FAQs

9.2 In-App Content

Chat Messages

- Welcome message
- Question prompts
- Clarification requests
- Error messages
- Completion message

Results Page

- Results header with summary
- Firm card content
- Regulatory status explanations
- Price disclaimer

Modals

- Appointment confirmation
- Callback request
- Save session
- Start new chat warning

10. Business Rules

10.1 Pricing Rules

- Prices displayed excluding VAT unless stated
- VAT calculated at 20% on legal fees (not disbursements)
- Disbursements shown as estimates where third-party
- "POA" (Price on Application) shown for complex scenarios
- Price valid for 30 days from quote generation

10.2 Enrollment Rules

- Only SRA-regulated firms can enroll
- Firms must accept CML terms to appear in top 5 results
- Non-enrolled firms shown in position 6+ without action buttons
- Firms can request removal from platform

10.3 Review Rules

- Minimum 10 characters for review text
- Rating required (1-5 stars)
- Reviews cannot be edited after 24 hours
- Firm responses limited to 500 characters
- Inappropriate content flagged for moderation
- CML reviews require valid appointment token

10.4 Data Retention

- Chat sessions: 30 days (extended if saved)
- Appointments: 7 years (legal requirement)
- Reviews: Indefinite
- Firm data: Until firm requests removal

10.5 Price Guarantee

- If firm does not honour quoted price, user contacts CML
- CML investigates and may remove firm from platform

- Displayed disclaimer: "Please ensure the firm confirms and honours the price you've seen here. If the firm does not honour the quoted price, contact us immediately - we'll step in to help."

Document Control

Version	Date	Author	Changes
1.0	December 2025	SOZO	Initial draft

— End of Functional Specification —

Phase 2 Feature List

SOZO

Phase 2 Feature List

Version: 1.0

Date: December 2025

Status: Draft

Prepared By: SOZO

Client: John Quentin / Choose My Lawyer Ltd

Overview

This document outlines potential features for Phase 2 development, building upon the MVP foundation. Each feature includes a description and the key requirements to implement it.

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1. Enhanced Firm Analytics Dashboard

Description

Provide deeper insights to enrolled firms about their performance on the platform beyond the basic stats in Phase 1.

Requirements

- Detailed metrics: search appearances, clicks, conversions, conversion rate over time
- Position tracking: average position in results, best/worst positions, distribution
- Trend analysis: week-over-week and month-over-month comparisons
- Geographic breakdown: searches and appointments by region
- Comparison benchmarks: performance relative to similar firms
- Visual charts for key metrics over time
- Data export functionality (CSV format)

2. Firm Self-Service Pricing Management

Description

Allow enrolled firms to update their own pricing data through the dashboard, reducing reliance on AI extraction.

Requirements

- Pricing editor interface matching the price card schema structure
- Support for band-based pricing, fixed fees, percentage-based pricing
- Adjustment/supplement configuration (e.g., leasehold supplement, new build premium)
- Disbursement list management
- Preview functionality showing calculated quotes with test inputs
- Version history for price changes
- Validation to ensure submitted pricing matches schema requirements
- Optional: approval workflow before changes go live

3. Motoring Offences Practice Area

Description

Expand the platform to cover summary motoring offences (e.g., speeding, drink driving, driving without insurance).

Requirements

- Question schema for motoring offences covering offence type, plea, court location, hearing expectations
- Price card schema for motoring matters
- Chat flow implementation for motoring enquiries
- WordPress landing page for SEO
- AI extraction pipeline for motoring transparency pages
- Initial dataset of extracted price cards from active firms

4. Employment Tribunals (Claimant) Practice Area

Description

Expand the platform to cover employment tribunal claims for individuals (unfair dismissal, wrongful dismissal, discrimination).

Requirements

- Question schema covering claim type, ACAS status, service length, salary, desired outcome
- Price card schema for employment tribunal matters
- Chat flow implementation for employment enquiries
- WordPress landing page for SEO
- AI extraction pipeline for employment transparency pages
- Initial dataset of extracted price cards from active firms

5. Immigration Practice Area

Description

Expand the platform to cover immigration services excluding asylum (spouse visas, work permits, ILR, citizenship).

Requirements

- Question schema covering application type, UK/overseas applicant, dependants, previous refusals
- Price card schema for immigration matters
- Chat flow implementation for immigration enquiries
- WordPress landing page for SEO
- AI extraction pipeline for immigration transparency pages
- Initial dataset of extracted price cards from active firms

6. Firm Profile Pages

Description

Public-facing profile pages for enrolled firms showing their details, services, pricing summary, and reviews.

Requirements

- Profile page displaying firm name, logo, offices, contact details
- Practice areas offered with pricing indication
- Aggregated review display with individual reviews
- Regulatory status and SRA information
- Link to firm's own website
- SEO optimisation for firm name searches
- Entry point to chat with pre-selected firm (if user wants to compare)

7. Admin Review Moderation Interface

Description

Enhanced administrative tools to manage and moderate user-submitted reviews beyond basic moderation.

Requirements

- Advanced filtering and search for moderation queue
- Bulk moderation actions
- Firm dispute handling workflow with escalation
- Automated flagging rules configuration
- Reporting dashboard for moderation metrics
- Full audit trail of all moderation actions

8. Email Marketing Integration

Description

Capture leads for remarketing to users who don't complete appointments.

Requirements

- Integration with email marketing platform (e.g., Mailchimp, HubSpot)
- Capture of email addresses from saved sessions
- Automated nurture sequences for abandoned sessions
- Unsubscribe handling and preference management
- GDPR-compliant consent collection

9. Multi-Lingual Support (Welsh)

Description

Provide Welsh language option for the chat interface to comply with Welsh Language Act requirements.

Requirements

- Translation of all chat questions and system messages
- Language selection option in interface
- WordPress content translation for Welsh visitors

10. Accessibility Enhancements

Description

Ensure the platform meets WCAG 2.1 Level AA accessibility standards.

Requirements

- Full keyboard navigation support throughout chat and dashboard

- Screen reader compatibility testing and fixes
- Colour contrast compliance (minimum 4.5:1 ratio)
- Focus indicators on all interactive elements
- Alternative text for all images
- Accessible form labels and error messages
- Skip navigation links

11. A/B Testing Framework

Description

Enable testing of different chat flows, UI variations, and messaging to optimise conversion.

Requirements

- Framework to define and run experiments
- Random user assignment to test variants
- Tracking of key metrics per variant (completion rate, appointment rate)
- Statistical significance calculation
- Admin interface to create, monitor, and conclude experiments

12. Firm Subscription Tiers

Description

Introduce paid subscription options for firms wanting enhanced features.

Requirements

- Definition of tier benefits (e.g., priority placement, enhanced analytics, API access)
- Payment integration (Stripe or similar)
- Subscription management in firm dashboard

- Automated billing and invoicing
- Grace period handling for failed payments

13. Partner API

Description

Provide API access for partners to integrate CML data into their own applications.

Requirements

- API key management for partners
- Rate-limited endpoints for organisation and pricing data
- Documentation (OpenAPI specification)
- Usage tracking and reporting
- Terms of service for API usage

14. Embeddable Widget for Law Firm Websites

Description

Allow law firms to embed a pricing calculator widget on their own websites that uses CML data.

Requirements

- JavaScript widget that can be embedded via script tag
- Configuration options (branding, colours, practice areas)
- Widget displays firm's own pricing based on user inputs
- Optional: lead capture that feeds back to CML
- Hosted widget infrastructure

15. Debt Recovery Practice Area (B2B)

Description

Expand to cover debt recovery services for businesses (up to £100,000).

Requirements

- Question schema covering debt amount, debtor type, dispute status, enforcement needs
- Price card schema for debt recovery matters
- Chat flow implementation for debt recovery enquiries
- WordPress landing page for SEO
- AI extraction pipeline for debt recovery transparency pages
- Initial dataset of extracted price cards from active firms

16. Employment Tribunals (Respondent) Practice Area (B2B)

Description

Expand to cover employment tribunal defence for businesses.

Requirements

- Question schema covering claim type, employee history, complexity, representation needs
- Price card schema for employment respondent matters
- Chat flow implementation for employment defence enquiries
- WordPress landing page for SEO
- AI extraction pipeline for employment respondent transparency pages
- Initial dataset of extracted price cards from active firms

17. Licensing Applications Practice Area (B2B)

Description

Expand to cover licensing applications for business premises.

Requirements

- Question schema covering application type, premises type, activities, location factors
- Price card schema for licensing matters
- Chat flow implementation for licensing enquiries
- WordPress landing page for SEO
- AI extraction pipeline for licensing transparency pages
- Initial dataset of extracted price cards from active firms

Document Control

Version	Date	Author	Changes
1.0	December 2025	SOZO	Initial draft

— End of Phase 2 Feature List —

Technical Specifications

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Technical Specifications

Table of Contents

1. Technology Stack Overview
2. System Architecture
3. Platform 1: WordPress Content Site
4. Platform 2: Custom Web Application
5. Data Architecture
6. AI/LLM Integration Strategy
7. External Integrations
8. Security Architecture
9. Performance & Scalability
10. Deployment Strategy

1. Technology Stack Overview

1.1 Two-Platform Approach

The solution is split into two platforms to optimise for different requirements:

Platform	Focus	Rationale
Platform 1 (Content)	SEO, content management, marketing flexibility	WordPress provides excellent SEO capabilities, familiar CMS experience, and vast plugin ecosystem
Platform 2 (Application)	Real-time interaction, complex business logic, AI integration	Custom-built application provides performance and flexibility for the comparison tool

1.2 Technology Choices

Frontend (Application)

Framework: React with Next.js

- Server-side rendering improves initial page load and SEO for application routes
- App Router provides excellent routing capabilities
- React's component model suits the chat interface well
- TypeScript adds type safety for a more maintainable codebase

Styling: Tailwind CSS

- Enables rapid development without custom CSS overhead
- Maintains consistency through utility classes

Backend (Application)

Framework: Python with FastAPI

- Python's excellent AI/ML ecosystem makes LLM API integration straightforward
- FastAPI provides high performance through async capabilities
- Automatic OpenAPI documentation

- Built-in validation through Pydantic

Database

Technology	Purpose
PostgreSQL	Primary database - robust JSON support, PostGIS extension for geolocation, proven reliability
Redis	Caching and session storage - speed ideal for real-time chat experience

WordPress

- Managed installation handles all content pages
- Standard plugins for SEO, caching, and forms
- Chat widget embedded via JavaScript from custom application

1.3 Infrastructure

Component	Technology	Purpose
Cloud Provider	Google Cloud Platform (GCP)	Hosting, compute, storage
Container Orchestration	Google Kubernetes Engine (GKE)	Production scaling, deployment
Local Development	Docker Compose	Environment consistency
CDN	CloudFlare	Caching, DDoS protection, SSL termination
Monitoring	Prometheus + Grafana	Metrics, alerting
Logging	GCP Cloud Logging	Centralised log aggregation

2. System Architecture

2.1 High-Level Overview

Users access the platform through a CDN/WAF layer that routes requests based on URL pattern:

Route Type	Platform	Content
Marketing site, landing pages, blog	WordPress (Kinsta)	Static/content pages
Chat interface, results, firm dashboard	Custom Application (GKE)	Dynamic functionality

Application Architecture: Standard three-tier

- **Frontend:** Next.js communicates with backend via REST APIs
- **Backend:** FastAPI handles business logic
- **Data Layer:** PostgreSQL (persistence) + Redis (caching)
- **Background Tasks:** Celery workers with Redis message broker

2.2 Data Flow Patterns

SRA Data Sync

1. Scheduled task runs daily
2. Fetches complete list of regulated organisations from SRA Data Share API
3. Compares incoming data with existing records
4. Actions performed:
    - Creates new organisations
    - Updates changed records
    - Flags removed organisations

Price Data Extraction

1. For each organisation with a website, scraper locates price transparency pages
2. Raw HTML/PDF/Markdown content is cleaned
3. Content sent to LLM for structured extraction
4. Extracted price card validated against schema and stored

5. Low-confidence extractions flagged for manual review

User Chat Flow

1. New session record created in database and cached in Redis
2. Session updated incrementally as user answers questions
3. On completion:
    - Price calculation engine queries matching organisations
    - Calculates quotes for each
    - Returns sorted results

Appointment Creation

1. Record persisted to database
2. Async tasks triggered for:
    - Email to user
    - Email to firm

Review Aggregation

1. Discover external profile IDs on firm enrollment
2. Scheduled sync fetches reviews from Google, Facebook, Trustpilot
3. Calculate aggregate ratings after each sync
4. Store reviews locally for persistent display

2.3 Integration Points

Service	Purpose	Phase
SRA Data Share API	Regulatory data	MVP
Claude API	LLM capabilities	MVP
Sparkpost	Transactional email	MVP
Postcode lookup (Fetchify.io	Address validation	MVP
Jina.ai	Scraping platforms	MVP
Google Places API	Review aggregation	MVP
Facebook Graph API	Review aggregation	MVP
Trustpilot API	Review aggregation	MVP

3. Platform 1: WordPress Content Site

3.1 Purpose and Scope

Primary Goals:

- Attract organic search traffic
- Educate potential users about the service
- Provide entry points into the comparison tool

3.2 Content Architecture

Section	Purpose
Homepage	Service introduction with prominent CTA to chat interface
Practice Area Landing Pag	SEO-optimised content (Conveyancing, Probate) with chat widget entry points
Knowledge Hub	Blog articles and guides for topical authority
For Law Firms	Benefits explanation and registration pathway

3.3 WordPress Configuration

Component	Choice	Purpose
Theme	Custom / Elementor Pro	Flexible, non-technical editing
SEO Plugin	Yoast SEO or RankMath	Meta, sitemaps, schema
Caching	WP Rocket	Performance
Forms	Gravity Forms	Contact, enquiries
Security	WordFence	WAF, scanning, login protection
Hosting	Kinsta	Managed WordPress, GCP infrastructure

3.4 Chat Widget Integration

- Embedded on WordPress pages via JavaScript snippet
- Widget loads in iframe for security isolation
- Communicates with parent page via postMessage API
- WordPress site remains stateless

4. Platform 2: Custom Web Application

4.1 Frontend Architecture

Route Groups

Group	Routes	Access
Public	Chat interface, results page, review submission	Anonymous
Authenticated	Firm dashboard	Firm users only

Key Components

Chat Interface (most complex):

- Real-time message state management
- Multiple input methods (button selection, text input, voice)
- Progress sidebar updates
- State management: React Context (session data) + React Query (server state)

Results Page:

- Sortable, filterable table of firm results
- Firm cards showing calculated quote, rating, action buttons
- Modals for appointment confirmation and callback requests

Firm Dashboard:

- Standard admin interface
- Navigation, data tables, forms, charts
- Review management with response functionality
- Authentication via HTTP-only cookies with JWT tokens

4.2 Backend Architecture

Module Organisation

Module	Purpose
Public API routes	Consumer functionality
Firm API routes	Authenticated firm features
Admin routes	Platform management
Background services	Data processing

Core Services

Service	Responsibility
ChatService	Session state and question flow management
PriceCalculator	Matches price cards to user answers
SearchService	Organisation queries with sorting/filtering
EmailService	Transactional email sending
LLMService	Interface to Claude API
ReviewService	Review aggregation, ratings, responses
ReviewSyncService	External platform review fetching

4.3 Authentication Flow

Consumer Users

- Anonymous (no authentication required)
- Sessions identified by UUID
- Persisted via URL for "save for later" feature

Firm Users

Token Type	Expiry	Storage	Purpose
Access Token (JWT)	1 hour	HTTP-only cookie	Request authentication
Refresh Token	7 days	HTTP-only cookie	Obtain new access tokens

4.4 Real-Time Considerations

Performance Optimisations:

- All chat API endpoints are async
- Database queries use connection pooling
- Session state cached in Redis to minimise DB round trips

5. Data Architecture

5.1 Core Entities

Entity	Description	Key Fields
Organisations	SRA-regulated law firms	SRA number, name, type, authorisation status, enrollment status, subscription tier
Offices	Physical locations	Address, geolocation coordinates
Price Cards	Structured pricing for practice area/matter type	Pricing model, adjustments, disbursements (JSON)
Chat Sessions	User's comparison journey state	Answered questions, message history, results (expires 30 days)
Appointments	User appointing firm or requesting callback	Monetisation trigger, user-firm link
Reviews	Reputation data from multiple sources	Source, rating, text, firm response
Review Source Config	External platform profile mappings	Google Place ID, Facebook Page ID, Trustpilot ID
Review Invitations	Secure tokens for CML native reviews	Token, appointment link, expiry
Firm Users	Dashboard authentication	Organisation, role, permissions

5.2 JSON Schema Approach

Why JSON for Price Cards:

- Different firms structure pricing differently:
  - Value bands
  - Percentages
  - Fixed fees
- Schema defines allowed structures while permitting variation
- Supports evolution without database migrations

5.3 Geolocation

- Office locations stored as PostGIS geography points
- Enables efficient distance calculations via spatial indexes
- User postcode lookup triggers proximity sorting

5.4 Data Lifecycle

Data Type	Retention Period	Notes
Session data	30 days default	Extended indefinitely if saved
Appointment data	7 years	Legal compliance requirement
Review data	Indefinite	Never deleted
Organisation meta data	Until removal request	Refreshed daily from SRA

6. AI/LLM Integration Strategy

6.1 Two Modes of AI Usage

Mode	Context	Priority
Price Extraction	Batch processing of transparency pages	Quality over speed
Chat Intelligence	Real-time user input processing	Speed (sub-second) and reliability

6.2 Price Extraction Pipeline

Process Flow:

1. Clean raw content (remove navigation, ads, scripts) → plain text
2. Send to LLM with detailed prompt containing:
   - Target schema
   - Extraction instructions
3. LLM identifies and returns structured JSON:
   - Pricing bands
   - Conditions
   - Adjustments
   - Disbursements
   - Scope information
4. Validate against schema
5. Assign confidence score

Confidence Handling:

Score	Action
< 0.7	Queue for manual review
≥ 0.7	Immediately available for calculations

6.3 Chat Intelligence

Primary Mode: Predefined questions with button selections (no AI required)

AI Enhancement Scenarios:

Scenario	Example	AI Action
Intent Recognition	"I'm buying my first house and need help"	Identify as conveyancing, route question flow
Entity Extraction	"about £350k" instead of precis number	Extract numeric value
Clarification	Ambiguous input	Generate contextual clarificatio question

Performance Requirements:

- Simple pattern matching handles common cases
- AI invoked only when rule-based logic fails
- Strict 2-second timeout to prevent chat blocking

6.4 Model Selection

Use Case	Model	Temperature	Rationale
Price Extraction	Claude Sonnet	0.1 (low)	Strong instruction-following, consistent outputs
Chat Intelligence	Claude Sonnet	0.1	Can substitute faster model if latency issues arise

6.5 Cost Management

Strategy	Implementation
Caching	Identical inputs return cached outputs
Batching	Extraction jobs run in bulk during off-peak hours
Fallbacks	Rule-based extraction handles simple cases without AI

7. External Integrations

7.1 SRA Data Share API

Purpose: Authoritative source for all regulated organisations in England and Wales

Sync Process (Daily):

1. Fetch all organisations
2. Compare against local records
3. Create new organisations
4. Update changed organisations
5. Flag removed organisations (not deleted - may have historical appointments)

Requirements:

- API key (obtained through registration)
- Rate limits generous for bulk operations

7.2 Email Service

Provider: Sparkpost

Features Used:

- Reliable delivery
- Bounce handling
- Analytics
- Template system (allows marketing to update copy without code changes)

Email Types:

- Appointment confirmations
- Callback request confirmations
- Session saved
- Review invitations (3-6 months after appointment)
- New review notifications to firms

7.3 Postcode/Address Lookup

Purpose: Validate postcodes and retrieve coordinates for distance calculations

Provider	Type	Coverage
Fetchify.io	Paid	Comprehensive

Usage:

- Called during chat flow when location question answered
- Results cached to avoid repeated API calls

7.4 Review Platform APIs

Purpose: Aggregate reputation data from external sources

Platform	API	Discovery Method	Sync Frequency
Google	Places API	Business name + address search	Every 6 hours
Facebook	Graph API	Page search	Every 6 hours
Trustpilot	Business API	Domain lookup	Every 6 hours

Process:

1. On firm enrollment, discover external profile IDs
2. Store mappings in review_source_config table
3. Scheduled task fetches new reviews
4. Reviews stored locally for display

7.5 Web Scraping

Provider: Jina.ai

Usage:

- Scraping law firm websites for price transparency pages
- Extracting raw content for LLM processing

8. Security Architecture

8.1 Authentication & Authorization

Consumer Users

- Unauthenticated
- Session identification via UUID tokens
- Acceptable as sessions contain no sensitive data beyond user inputs

Firm Users

- Email/password authentication with JWTs
- Access token contains: user ID, organisation ID, role
- Backend validates token and checks role permissions per request

8.2 Data Protection

Layer	Protection
Transport	HTTPS with TLS 1.3
Certificates	Managed via CDN or Let's Encrypt
Database connections	SSL
Data at rest	GCP encryption
Passwords	bcrypt (cost factor 12)
Secrets	GCP Secret Manager (never in version control)

8.3 Input Validation

- All API inputs validated using Pydantic schemas
- Invalid inputs rejected with clear error messages
- Prevents SQL injection, XSS, and other injection attacks
- User-generated content (reviews, firm responses) sanitised before display

8.4 Rate Limiting

Endpoint Type	Limit
Public API	100 requests/minute/IP

Endpoint Type	Limit
Authenticated	300 requests/minute/user
Login	5 attempts/minute (brute-force protection)
Review submission	3/hour per token

Implementation: CDN level (CloudFlare) with additional application-level limits

8.5 GDPR Compliance

Data Collection Principles:

- Collect only when necessary
- Store only as long as needed
- Support user deletion requests
- Appointments retained for legal compliance (can be anonymised after retention period)

Cookie Handling:

Platform	Approach
WordPress	Cookie consent implemented
Application	Essential cookies only (session, authentication)

8.6 WordPress Security

The WordPress site is protected using WordFence Security, which provides:

- **Web Application Firewall (WAF)**: Blocks malicious traffic before it reaches WordPress, protecting against SQL injection, XSS, and other common attacks
- **Malware Scanning**: Regular scans of core files, themes, and plugins to detect compromised code
- **Login Security**: Brute force protection, two-factor authentication, and CAPTCHA support
- **Real-time Threat Intelligence**: Updates from WordFence's threat defence feed

The managed WordPress host Kinsta provides additional infrastructure-level security including DDoS protection, automatic security patches, and isolated container environments.

9. Performance & Scalability

9.1 Performance Targets

Metric	Target	Measurement
Page load	< 2 seconds	Time to First Contentful Paint
API response	< 200ms (p95)	Excluding LLM calls
Chat message	< 1 second	Including AI processing
Price calculation	< 500ms	100+ firms comparison

9.2 Caching Strategy

Layer	Technology	TTL	Invalidation
Static assets	CloudFlare CDN	1 year	Content hash
Organisation data	Redis	24 hours	On SRA sync
Price cards	Redis	24 hours	On update
Chat sessions	Redis	30 days	On expiry

9.3 Database Performance

Optimisations:

- Indexes on all frequently queried columns
- PostGIS GIST spatial index for location queries
- Mandatory pagination for list endpoints
- Read replicas for analytics/reporting queries
- SQLAlchemy async connection pooling

9.4 Horizontal Scaling

Component	Scaling Method
Frontend	Add container instances (GKE)
Backend	Add container instances (GKE)
Database	Vertical + read replicas

Component	Scaling Method
Redis	Clustering if needed
Celery workers	Add worker instances

The application is stateless—any request can be handled by any instance. This enables horizontal scaling by adding container instances behind the load balancer.

9.5 Monitoring & Alerting

Tool	Purpose
Prometheus	Metrics collection
Grafana	Visualisation, dashboards
GCP Cloud Monitoring	Infrastructure metrics
GCP Cloud Logging	Centralised logs

Alerts trigger for error rate spikes, latency increases, or resource exhaustion.

10. Deployment Strategy

10.1 Environment Progression

Environment	Infrastructure	Purpose
Development	Local Docker Compose	Developer workstations
Staging	GKE (smaller cluster)	Pre-production testing
Production	GKE (full cluster)	Live users

All environments use the same container images, with configuration provided via environment variables.

10.2 CI/CD Pipeline

Build Stage (Push to main):

1. Code linting
2. Unit tests
3. Integration tests
4. Docker image build
5. Push to Google Container Registry

Deploy Stage:

- **Staging:** Auto-deploy on merge to release branch
- **Production:** Manual approval required

10.3 Database Migrations

Aspect	Approach
Tool	Alembic
Testing	Staging before production
Compatibility	Backwards-compatible only
Rollback	Rollback scripts included

Migrations must be backwards-compatible to enable zero-downtime deployments.

10.4 Rollback Strategy

Scenario	Action	Time
Container issue	Revert to previous image	Minutes
Database issue	Run rollback migration	Minutes
Feature issue	Disable via feature flag	Seconds

10.5 WordPress Deployment

Aspect	Approach
Hosting	Kinsta managed WordPress
Content updates	Direct in WordPress admin
Theme/plugin updates	Staging → Production
Backups	Automatic (Kinsta)

Document Control

Version	Date	Author	Changes
1.0	December 2025	SOZO	Initial draft

Costs Estimate

SOZO

Cost Estimate

The following estimates are subject to change following agreement of final Technical Specifications Document.

Fullstack Development

- Includes Frontend, backend, Data & Infrastructure, AI Integration

£80K - £120K

Project Management

£8K - £12K

Per Additional Practice Area

- Complex (Immigration, Employment Tribunals): **£10k - £12k**
- Basic (Wills, Motoring Offences): **£5k - £7k**

Design Updates

Updates may be required following feedback from stakeholders/investors

£2K - £5K